ASCE 7-22 Seismic Screening

Design response spectrum (SDS, SD1), site class sensitivity, and base shear from USGS.

Screening only — not for final design. Verify with site-specific studies and ASCE 7-22.

Parameters

Location, Risk Category, and Site Class Inputs

Location

Los Angeles, CA

Site Class

Default

Risk Category ℹ

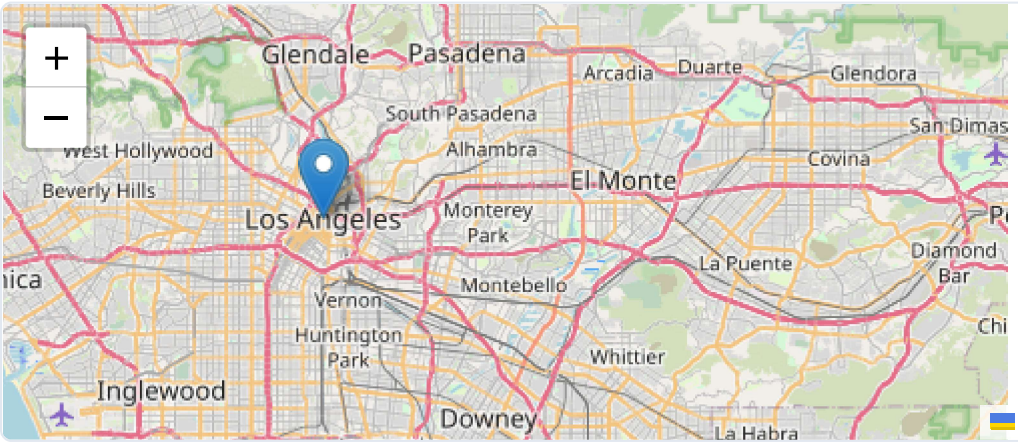
I II III IV

Affects Ie and SDC thresholds

Sensitivity Chart ℹ

Simplified Detailed SDS SD1

Simplified: A, B, C, D, E · Detailed: Default + full list



34.052200, -118.243700 Copy Open in Google Maps

Key Values

Last updated 10:15 PM

SDS	SD1	SDC
1.580 g	**1.110 g**	**D**
Short-period design spectral acceleration	1-sec design spectral acceleration	Seismic Design Category
Δ +0.480g	Δ -0.120g	

Ie	Ts	TO
1	**0.701 s**	**0.140 s**
Importance factor	Transition period	Corner period

Δ -0.419s

Δ -0.084s

TL ⓘ

8 s

Long-period transition

Δ -8.000s

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Base Shear (ELF) Calculator

Screening-level equivalent lateral force using ASCE 7-22

Response modification coefficient (R) ⓘ

8

Effective seismic weight (W, kips) ⓘ

e.g. 5000

Fundamental period (T, s) ⓘ

e.g. 0.5

S1 (MCEr), g override ⓘ

0.720

Importance factor (Ie) ⓘ

1

☐ Override (from Risk Cat.)

Enter W (effective seismic weight, kips) and T (fundamental period, s) to compute base shear. Values must be within valid ranges.

Screening only. Final ASCE 7-22 ELF requires additional checks (period limits, system selection, vertical distribution, etc.).

Risk Category Compare

RC II vs RC IV — Los Angeles, CA, Site Class Default

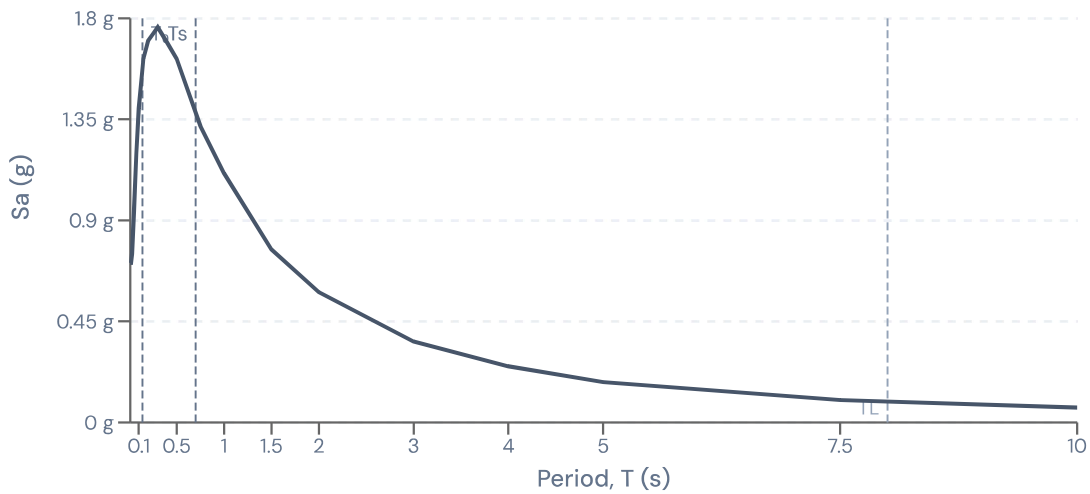
RISK CATEGORY II		RISK CATEGORY IV	
Ie **1.0**	SDC **D**	Ie **1.5**	SDC **D**
SDS **1.580g**	SD1 **1.110g**	SDS **1.580g (unchanged)**	SD1 **1.110g (unchanged)**

Ie and SDC vary with Risk Category. SDS and SD1 typically remain unchanged (USGS hazard by location + site class).

Design Response Spectrum ⓘ

Design spectral acceleration (Sa) vs. period — Los Angeles, CA, Site Class Default

Design MCEr Log Linear



Site Class Sensitivity (SDS)

SDS by site class — Los Angeles, CA Reference: D

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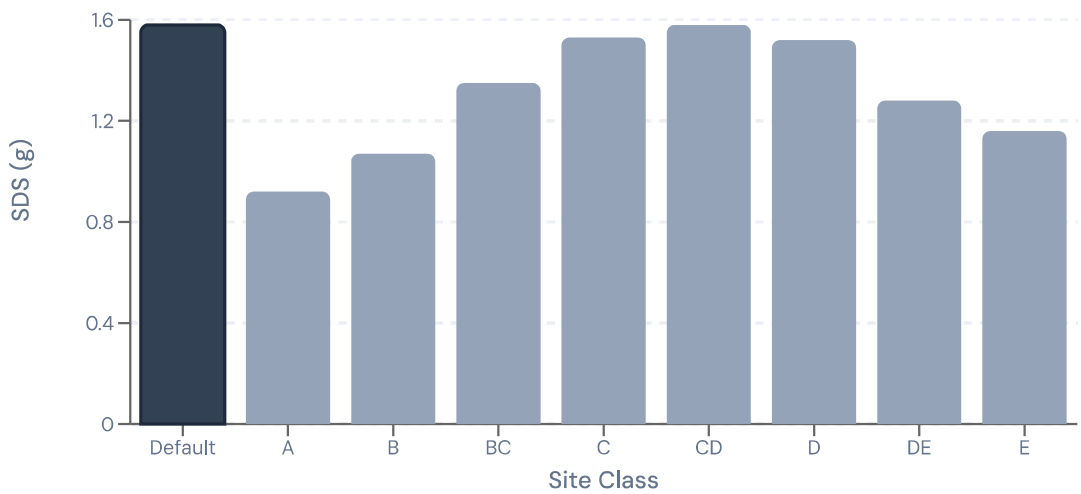

Interpretation

Summary, Key Parameters, and Next Steps for Los Angeles, CA

SUMMARY

Short-period design spectral acceleration 1.580g

KEY PARAMETERS

Primary design values SDS = 1.580g, SD1 = 1.110g

NEXT STEPS

- Use SDS and SD1 with your R factor and Ie for base shear

1-sec design spectral acceleration	1.110g	Spectral corners	T0 = 0.140s, Ts = 0.701s, TL = 8.000s
Seismic Design Category	D	Highest SDS site class	Default (1.580g, +3.9% vs D)
Importance factor (Risk Cat. affects Ie & SDC thresholds)	1		

calculation.

- Risk Category mainly affects Ie and SDC; SDS/SD1 depend on location and site class.
- Higher SDC may require additional detailing per ASCE 7-22.